# ULTRAVIOLET (UV) INDEX

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Ultraviolet (UV) Index, UVI, was developed in 1994 by the Environmental Protection Agency (EPA) and National Weather Service (NWS) to help the general public make informed decisions and protect themselves when engaging in outdoor activities. UVI measures the level of UV radiation expected on a given day. The daily UVI is important for the public to be aware of since increased exposure to UV radiation increases one's risk for the lifetime development of skin cancer.

The reported UVI predicts the UV level at solar noon, which typically occurs between 12pm and 2pm. The UVI varies depending on latitude, altitude, cloud cover, ozone, season, earth surface characteristics, land cover, and time of day. The UVI also rises and falls throughout the day, however the highest daily value is what gets reported [Figure 1].

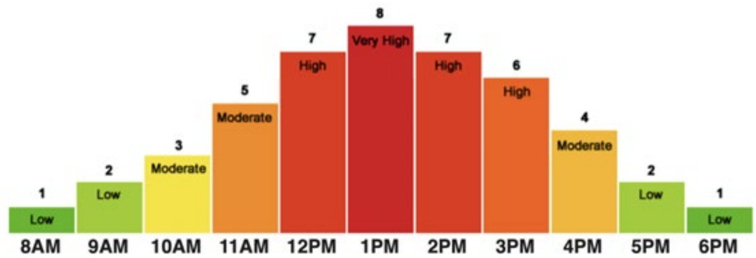


**Figure 1.** Example of UV index (UVI) variation on a summer day in Washington DC (EPA.gov).

In 2002 the World Health Organization (WHO), the World Meteorological Organization, and the United Nations Environment Programme and the International Commission on Non-Ionizing Radiation Protection recommended the Global Solar UV Index to make UV reporting consistent worldwide. These recommendations included a new scale and color scheme.

As of 2004, the EPA and NWS adopted the Global Solar UV Index guidelines recommended in 2002. The UVI reported values range from 1 to 11+, where 1 is equivalent to exposure category LOW and 11+ is equivalent to exposure category EXTREME. The higher the UVI, the greater and faster damage can occur to the skin and eyes.

| Exposure Category | Index Number | Sun Protection Messages |
|---|---|---|
| **LOW** | 1-2 | -Wear sunglasses on bright days.<br>-If you are prone to burning, apply sunscreen and cover up. |
| **MODERATE** | 3-5 | -Apply sunscreen and cover up.<br>-Find shade during the middle of the day. |
| **HIGH** | 6-7 | **Protect your skin from a sunburn by:**<br>-Applying sunscreen, covering up, wearing a hat and sunglasses.<br>-Minimizing outdoor sun exposure between 11am-4pm. |
| **VERY HIGH** | 8-10 | -Extra precautions needed, as sunburns can occur quickly.<br>-Avoid outdoor sun exposure between 11am-4pm. If necessary, seek shade, apply sunscreen, cover up, wear a hat and sunglasses. |
| **EXTREME** | 11+ | -Take **ALL** precautions.<br>-Sunburns can occur within minutes.<br>-Avoid outdoor sun exposure between 11am-4pm. If necessary, seek shade, apply sunscreen, cover up, wear a hat and sunglasses. |

**Table 1.** Ultraviolet index (UVI) number with associated exposure category and recommended corresponding sun protection measures.

**This information has been provided to you compliments of the American Osteopathic College of Dermatology and your physician.**

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The EPA recommends the following sun protection tips, regardless of reported UVI:

- Check the UV Index
- Limit sun exposure during the middle of the day
- Seek shade
- Wear clothing with tightly woven fabrics to prevent UV rays from penetrating through (ie. long sleeve shirts and pants)
- Wear a wide brimmed hat
- Wear sunglasses with UV protection
- Use broad-spectrum SPF and reapply throughout the day

The best protection from the sun includes hats, shade, and clothing.

To find out what the UV Index is today in your area visit the EPA Website: **https://www.epa.gov/sunsafety/uv-index-1**

**References:**


**https://www.epa.gov/sites/default/files/documents/uviguide.pdf**

**https://www.who.int/news-room/questions-and-answers/item/radiation-the-ultraviolet-(uv)-index**

**https://www.weather.gov/ilx/uv-index**